

CHAPTER ONE
INTRODUCTION TO STATISTICS
DEFINITION OF STATISTICS

The word statistics has two meanings. That is we can define Statistics in two senses.

Statistics defined in plural sense (As a data)

In this sense, it is equivalent to referring to numerical facts, figures or statistical data. i.e., the raw data themselves, like statistics of births, statistics of students, statistics of imports and exports, etc.

But not all numerical data are statistics.

Some of the characteristics which numerical data must possess in order that they may be called statistics are given below.

1. Statistics should be numerically expressed
2. They should be aggregates of facts
3. They should be collected in a systematic manner
4. They should be collected for a predetermined purpose
5. They should be placed in relation to each other
6. They should be enumerated or estimated according to reasonable standards of accuracy

In general it can be said that all statistics are numerical facts; but not all numerical facts are statistics.

In order numerical facts to be statistics, the above characteristics have to be satisfied.

Statistics defined as a method (Singular Sense)

The second meaning of statistics refers to the science or discipline of study. In this sense of the word, Statistics is defined as the science of collecting, presenting, analyzing and interpreting numerical data to make decisions.

BASIC TERMINOLOGIES IN STATISTICS

Statistics has its own terms and words. Knowing these terms and words is fundamental in understanding the Statistical methods and concepts. Some of the basic terms that will be used throughout the course are defined below.

- a) **Data:** In Statistics, all conclusions are based on facts and the first step in any statistical investigation is to collect a set of related observations from which conclusions may be drawn. Such related observations that form the set are known as Data. The word data was obtained from the singular Latin word “datum “to mean fact.

- b) **Population:** The complete collection of individuals, objects, or measurements that have a characteristic in common or totality of related observations in a given study is described as a population. The population that is being studied is also called the **target population**.

Example 1.1

- i) Population of trees underspecified climatic conditions.
- ii) Population of animals fed a certain type of diet.
- iii) Population of farms having a certain type of natural fertility.
- iv) Population of households, etc. Population can be finite (limited in its size) or infinite (unrestricted).

c) **Sample:** A sub group of the population that will be studied in detail is called a Sample.

Example 1.2 In studying the average marks of students in Stat 181, if you take marks of 20 students out of all 150 students who might have taken the course, the collected data will be a sample.

d) **Parameters:** are statistical measures obtained from a population data. These measures may include the mean, variance, standard deviation, etc. and are denoted by μ, σ^2, σ , etc. respectively.

- e) **Sample Statistic:** a number computed from a sample data. Sample Statistics are denoted by lower case letters of the alphabet such as $\bar{x}$ -the sample mean, s^2 - the sample variance, etc
- f) **Variable:** is a characteristic under study that assumes different values for different elements and most of the time variables are denoted by the letters X, Y, Z, etc.

Example 1.3: Height, Weight, Age, Income, Expenditure, Grade, Intelligence, sex, color, etc.

- g) **Quantitative Variable:** is variable that can be expressed numerically such as height, weight, age, income, expenditure, grade, family size, number of students in a class, etc.
- h) **Qualitative Variable:** is variable that cannot assume a numerical value but can be classified into two or more nonnumeric categories such as the gender of a person, the language in which a book is written, hair color, and so on.
- i) **Discrete Variable:** is a variable whose values are countable such as family size, number of students in a class, etc. Its values are obtained by counting.
- j) **Continuous Variable:** is a variable which can, theoretically assume any numerical value between two given values. The respective values of such continuous variables are obtained by measuring such as time taken to complete a certain examination, height of a person, age of a person, test scores of a student and so on.
- k) **Observation or Measurement:** The value of a variable for an element is called an Observation or Measurement.

- l) **Data Set:** A data set is a collection of observations on one or more variables.
- m) **Sample Frame:** A list of the entire population from which items can be selected to form a sample is referred to as sample frame.

CLASSIFICATION OF STATISTICS

Statistical methods are classified into two groups or areas based on how data are used. These are: Descriptive Statistics and Inferential Statistics.

Descriptive Statistics consists of the collection, organization, presentation and analysis of numerical data. It is concerned with describing certain characteristics of a set of data (usually a sample) – that is, what it is shaped like, what number the values tend to cluster (converge) around, how much variation is present in the data, and so forth.

In short, Descriptive Statistics describes the nature or characteristics of data without making conclusion or generalization.

Example 1.4: The average age of athletes participated in London Marathon was 25 years, 80% of the instructors in Arsi University are males, The marks of 50 students in Statistics course are found to range from 50 to 85, etc. are some examples of Descriptive Statistics.

Inferential Statistics is concerned with the process of drawing conclusions (inferences) about specific characteristics of a population based on information obtained from samples, performing hypothesis testing, determining relationships among variables, and making predictions.

Example 1.5: The result obtained from the analysis of the income of 100 randomly selected citizens in Ethiopia suggests that the average daily income of a citizen in Ethiopia is 30 Birr i.e. the average income of all families in Ethiopia can be estimated from figures obtained from a few hundred families.

STEPS OF STATISTICAL INVESTIGATION

There are five steps of statistical investigation on numerical data. The indicated steps are discussed below

1. Collection of data

This is the process of obtaining measurements or counts and constitutes the first step in statistical investigation.

In general information pertinent to the underlying investigation are collected. Valid conclusions can only result from properly collected data. i.e., if data are faulty, the conclusions drawn can never be reliable. Hence, utmost care must be exercised in collecting data because they form the foundation of statistical analysis

2. Organization of data

Collected data have to be organized in a suitable form so that one can have a general understanding of the information gathered. A large mass of figures that are collected from surveys frequently need organization. The first step in organizing a group of data is editing. The collected data must be edited very carefully so that omissions, inconsistencies, irrelevant answers, wrong computation in the returns from a survey may be corrected or adjusted

After data are edited, the next step is to classify them. The purpose of data classification is to arrange them according to some common Characteristics possessed by the items constituting the data. The last step in data organization is tabulation. The purpose of tabulation is to arrange the data in columns and rows so that there is absolute clarity in the data presented.

3. Presentation

After the data have been collected and organized, they are ready for presentation. The main purpose of data presentation is to facilitate statistical analysis. This can be done by arranging the data using graphs and diagrams.

4. Analysis of Data

After collection, organization and presentation the next step is that of analysis. This is the extraction of summarized and comprehensible numerical descriptions of the data where these measures will in turn give a far better understanding the nature of the data.

The purpose of analyzing data is to dig out information useful for decision making

Methods used in analyzing the presented data are numerous, ranging from simple observation of the data to complicated, sophisticated and highly mathematical techniques. The most commonly used methods of statistical analysis are measures of central tendency, measures of variation, correlation, regression, estimation and hypothesis testing

5. Interpretation of data

This is the last stage in statistical investigation. It is the task of drawing conclusions from the analysis of the data and usually involves the formulation of predications concerning a large collection of objects from information available for a small collection of similar objects.

This step usually involves decision making about a large collection of objects (Population) and about information gathered from a small collection of similar objects (sample). The interpretation of data is a difficult task and necessitates a high degree of skill and experience.

APPLICATIONS AND LIMITATIONS OF STATISTICS

The study of statistics has become more popular than ever during the past three decades or so. The increasing availability of computers and statistical software packages has enlarged the role of statistics as a tool for empirical research. As a result, statistics is used for research in almost all professions. So the various tools of statistics are being used to solve problems in everyday life, in research, in marketing, in planning, in production and quality control, and other areas.

LIMITATION OF STATISTICS

Even though statistics is growing in popularity and in being successfully employed by the seekers of truth in numerous fields of learning, still it has limitations.

The field of statistics, though widely used in all areas of human knowledge and widely applied in a variety of disciplines such as business, economics and research, has its own limitations. Some of these limitations are

1. It does not deal with individual values. Statistics only deals with aggregate of values. For example, the mark obtained by one student in a class does not carry any meaning in itself unless it can be compared with a set standard or with other students in the same class or with their own marks obtained earlier.
2. It cannot deal with qualitative characteristics
3. Statistical conclusions are not universally true
4. Statistical interpretation requires a high degree of skill and understanding of the subject. In order to get meaningful results, it is necessary that the data be properly and professionally collected and critically interpreted. It requires intensive training to read and analyse statistics in its proper context. It may lead to fallacious conclusions in the hands of the inexperienced

USES OF STATISTICS

In general, statistical methods and procedures are useful almost in every one's life. The decision making process by the managers of modern business and industry is governed by statistical applications. So, statistical methods are applied to, virtually, any kind of situation where numerical information is gathered with the objective of making rational decisions in the face of uncertainty. Some of the uses of statistics are:

To present facts in a definite form

That is to present huge numerical information in a concise, unambiguous and easily understandable way and this helps proper comprehension of what is stated.

Statistics facilitates comparisons

Unless figures are compared with the others of the same kind, they are often devoid of any meaning. Statistics is a technique of making comparisons, using statistical measures like correlation, averages, coefficients, rates and ratio, etc. comparisons between two or more sets of data can be made to understand their relationship and differences

Statistics gives guidance in the formulation of suitable policies

From the past recorded statistics predication can be made to the future situations and accordingly policy makers could formulate their production program in the light of changes that are anticipated to take place in the quantity of sales.

For example, demographic data about population size, its distribution by age and sex and other socioeconomic characteristics, the rate of population growth, migration, death rate, etc. help policy makers in determining the future needs such as food, clothing, housing, education, health facilities, recreational places, water, electricity, transportation system, etc

Prediction

Plans and policies are usually formulated well in advance of time of their implementation; Acknowledge of future trends is very helpful in framing suitable policies and plans. Statistical methods provide helpful means of forecasting future events.

Statistical methods are very helpful in formulating and testing hypothesis and to develop new theories

Statistics in the sciences: The ideas and methods of statistics have been useful to the sciences- in the conduct of research in the analysis of findings and also in the formulation of laws.

SCALES OF MEASUREMENT

Dear students, If Melaku's ID.NO is R/008/2000 and Kebede's ID.NO is R/010/200, whom do you think is better student? Suppose again Melaku's mark in Statistics is 90 and Kebede's mark in Statistics is 80, whom do you think is better? Scales of measurement concern with such problems. Let us see them.

Measurement can be defined as the assignment of numbers to objects and events according to logically acceptable rules. The number system is highly logical and offers a multiplicity of possibilities of further logical manipulations.

A measurement scale should possess the following attributes to allow for these logical manipulations

Magnitude

Is the quantum or quantity in which the attribute exists in various instances of the phenomena. It allows us to tell whether one instance of the attribute is greater than less than or equal to another instance of the attribute.

Example 1.6: If X gets a score of 20 on an aggressiveness scale and Y a score of 25, we can say that Y is more aggressive than X.

Equal intervals

It denotes that the magnitude of the attribute represented by a unit of measurement on the scale is equal regardless of where on the scale the unit falls.

Example 1.7: A difference in heights between 60 inches and 65 inches is equal to the difference in height, between 67 inches and 72 inches.

Absolute zero point

Is a value that indicates exists at that point or nothing at all of the attribute being measured exists.

For example, a zero weight indicates “no weight” at all

Keeping in mind these three characteristics of measurements scales of measurement can be divided in to four different types.

1. Nominal Scale (Classificatory Scale)

It refers to the simple classification of objects or items in to discrete groups which do not bear any magnitude relationships to one another.

In the nominal scale of measurement, numbers are used simply as labels for groups or classes.

Example 1.8: If our data set consists of blue, green, and red items, we may designate blue as 1, green as 2, and red as 3. In this case, the numbers 1, 2 and 3 stand only for the category to which a data point belongs.

“Nominal” stands for “name” of category. The nominal scale of measurement is used for qualitative rather than quantitative data: blue, green, red, male, female; marital status (married, single, divorced, etc.); professional classification; geographic classification; and so on.

2. Ordinal Scale or Ranking Scale

First it is nominal scale and here data elements may be ordered according to their relative size or quality. It means that ordinal scale is first of all, nominal but most people would agree with the order in which the categories were placed so, in ordinal scale inequalities have a meaning the inequality signs '<' or '>' may assume any meaning like "stronger than", "softer than", "weaker than" etc.

Ordinal scale reflects only magnitude and does not possess the attribute of equal intervals or an absolute zero point

Example 1.9: Let us consider the set of all diplomas. B.Sc /B.A., and M.Sc. /MA graduates of a certain university at the end of a certain academic year. The graduates could be classified according to their level of education. It is possible to say "the M.Sc/MA graduate is better than the B.Sc. /BA. Graduate" or to say "the B.Sc./BA. Graduate is better than the diploma graduate". Therefore we can have meaningful inequalities.

3. Interval Scale

The interval scale possesses two out of three important requirements of good measurement scale i.e., magnitude and equal intervals but lacks the real or absolute zero point.

An interval scale is one, which provides equal intervals from an arbitrary origin. An interval scale not only orders according to the amount of the attribute they represent, but also establishes equal intervals between the units of measure. Equal differences in the numbers represent equal differences in the attribute being measured.

Example 1.10: If we apply a meter scale to measure heights of the students, we may find their height to be 150cm, and so on. These measurements are on a scale with equal intervals. The distance of height between 150cm and 160cm is exactly equal to the distance of height between 170cm and 180cm.

Example 1.11: The Fahrenheit and centigrade thermometers are examples of interval scales.

On an interval scale, both the order and distance relationships among the numbers have meaning. They may assert 30 and 31 degrees centigrade equal to the differences between 40 and 41 degrees centigrade. We could not say. However 50°C is twice as hot as 25°C this is because there is not true zero point on an interval scale. Since zero is arbitrary, multiplication and division of numbers are not appropriate.

4. Ratio Scale

The scale of measurement which has all the three attributes – magnitude, equal intervals and an absolute zero point- is called a ratio scale addition, subtraction, multiplication and division of the numbers are appropriate

Example 1.12:- All physical measurements, like height, weight, etc.

- Number of students in various classes.
- Number of books possessed by students of a class, etc.

CHAPTER TWO

METHODS OF DATA COLLECTION AND PRESENTATION

INTRODUCTION

Dear Students, do you know the types of data we usually use in statistical data analysis? If you know it is fine. The first step of a statistical study is gathering the necessary facts or data from the possible sources. They are the foundation of a statistics study or result. Thus, the method of data collection must be well designed and administered.

Statistical Data

Statistical data are the basic input needed to make an effective decision in a particular situation. The main reasons for collecting data are:

- i) To provide necessary inputs to a given phenomenon or situation under study.

- ii) To measure performance in an ongoing process such as production, service, and so on.
- iii) To enhance the quality of decision – making by enumerating alternative courses of action in a decision-making process, and selecting an appropriate one.
- iv) To satisfy the desire to understand an unknown phenomenon.
- v) To assist in guessing the causes and probable effects of certain characteristics in given situations.

Types of Data

Statistical data are the outcome of a continuous process of measuring, counting, and/or observing. These may pertain to several aspects of a phenomenon (or a problem) which are measurable, quantifiable, countable or classifiable. The researcher will collect and analyze data about the characteristics of the given population. These characteristics which one intends to investigate and analyze are termed as variables. Variables have two natures: those which are expressed in numerical terms and those which are not expressed in numerical terms. While sex, religion, and language are a few examples of non-numerical variables, age, weight, height, and distance are examples of numerical variables. The numerical variables are classified into two categories:

- i) **Discrete variables** – which can only take certain, fixed integer numerical values. For example, the number of cards or the numbers of employees in an organization are examples of discrete variables.
- ii) **Continuous variables** – This can take any numerical value. Measurements of height, weight, length, in centimeters/ inches, grams/kilograms are a few examples of continuous variables.

Remark: Discrete data numerical measurements arise from a process of counting; while continuous data are numerical measurements arise from a process of measuring.

Sources of Data

The choice of a data collection method from a particular source depends on the facilities available, the extent of accuracy required in analysis, the expertise of the investigator, the time span of the study, and the amount of money and other resources required for data collection.

Data sources are classified as primary sources and secondary sources.

i) Primary sources

Individuals, focus groups, and/or panels of respondents specifically decided up on and set up by the investigator for data collection are examples of primary data sources. Any one or a combination of the following methods can be chosen to collect primary data:

Observation: In observation studies, the investigator does not ask questions to seek clarifications on certain issues. Instead he/she records the behavior, as it occurs of an event in which he interested. Sometimes mechanical devices such as camera, tape recorders are also used to record the desired data.

Interviewing: Interviews can be conducted either face –to-face or over telephone. Such interviews provide an opportunity to establish a rapport with the interviewee and help extract valuable information. Direct

interviews are expensive and time-consuming if a big sample of respondents is to be personally interviewed. Interviewers' biases also come in the way. Such interviews should be conducted at the exploratory stages of research to handle concepts and situational factors.

Questionnaire: It is a formalized set of questions for extracting information from the target respondents. The form of the questions should correspond to the form of the required information. The three general forms of questions are: dichotomous (yes/no response type); multiple choice, and open-ended. A questionnaire can be administered personally or mailed to the respondents. It is an efficient method of collecting primary data when the investigator knows what exactly is required and how to measure such variables of interest.

ii) Secondary Data Sources

Secondary data refer to those data which have been collected earlier for some purposes other than the analysis currently being undertaken. There are external and internal sources of secondary data.

External secondary data sources: the external secondary data sources include government publications, non-government publications, various syndicate services such as Operations Research Group (ORG) and international organizations.

Internal secondary data sources: The data generated within an organization in the process of routine business activities, are referred to as internal secondary data. Financial accounts, production quality control, and sales records are examples of such data.

SCOPE OF THE DATA COLLECTION

At the beginning from the mission of the given study the investigator must identify the population and the particular variables or characteristics of the population on which data are required. Following the identification of the population and the variables of the study the investigator must decide up on the scope or coverage of the data collection. This to mean that he/she must decide whether data are to be collected from all units of the population or partial survey of items (sample) is sufficient.

CENSUS OR COMPLETE ENUMERATION

Census is a method of studying the whole population. Under the census or complete enumeration survey method, data are collected for each and every unit of the population or universe.

1. Data are obtained from each and every unit of the population
2. The results obtained are likely to be more representative, accurate and reliable
3. It is an appropriate method of obtaining information on rare events.

4. Data of complete enumeration (census) can be used as a basis for various surveys.

Despite these advantages the census method is not very popularly used in practice.

SAMPLE SURVEY

It is the process of collecting data covering a representative part of a population. That is the process of learning about the population on the basis of a sample drawn from it. Thus, in the sampling technique instead of every unit of the universe only part of the universe is studied and the conclusions are drawn on the basis for the entire universe.

Some of the advantages of sample survey are

1. It reduces cost: It saves time, costs less, and requires fewer skilled labors.
2. Greater speed: Data obtained from a sample takes less time to collect, process and to have the data published in time
3. Greater scope
4. Greater Accuracy
5. If the nature of the investigation is destructive or if the population is infinitely many, sample survey is the only practical way since census leads to disastrous results, or is impossible to conduct.

Therefore, sample surveys are more often conducted than census due to the above stated advantages. It must be noted, however, that the advantages could be enjoyed only if an appropriate sampling technique is used and the sample size is adequate.

ORGANIZATION AND PRESENTATION OF DATA

Dear learner, could you please mention some of the methods to organize and present the statistical data?

The best way to examine a large set of numerical data is first organize and present it in an appropriate tabular and graphical format. As the number of observations gets large, it becomes more and more difficult to focus on the specific features in a set of data. Thus we need to organize the observation so that we can better understand the information that the data are revealing.

The raw data can be organized in a data array and frequency distribution. Such an arrangement enables us to see quickly some of the characteristics of the data we have collected. When a raw data set is arranged in rank order, from the smallest to the largest observation or vice-versa, the ordered sequence obtained is called an ordered array.

FREQUENCY DISTRIBUTION

Definition: The number of times the values of a variable in a data occurs is called the frequency of a value.

Definition: A tabular representation of values of a variable together with the corresponding frequencies is called a frequency distribution (FD)

Types of frequency distributions

1. Ungrouped frequency distribution (UFD)

Shows a distribution where the values of a variable are linked with respective frequencies. Discrete frequency distribution is one which involves a discrete variable

Example 2.1: The following data represents the number of books read in the past six months by each student in a class of 25.

6	24	14	11	33
15	15	8	14	10
8	27	15	6	20
20	9	33	15	10
6	11	20	8	6

Required: Construct a discrete frequency distribution for this data.

Steps in constructing discrete frequency distribution

1. make sure that the variable you have is discrete
2. Determine the possible values of the variable
3. Prepare an array for the distribution of the variable
4. Prepare three columns, the first for the different values of the variable, the second for tally marks to facilitate the counting, the third for the frequency corresponding to each value of the variable

Values	Tally Marks	Frequency
6	1111	4
8	111	3
9	1	1
10	11	2
11	11	2
14	11	2
15	1111	4
20	111	3
24	1	1
27	1	1
33	11	2

5. Write the possible values of the variable in ascending order in the first column.

2. Categorical frequency Distribution

Used for data that can be placed in to specific categories such as nominal or ordinal e.g. marital status.

Example 2.2: A social worker collected the following data on marital status for 25 persons.

(M= married, S = Single, W= widowed, D= divorced)

M	S	D	W	D
S	S	M	M	M
W	D	S	M	M
W	D	D	S	S
S	W	W	D	D

Construct a frequency distribution for the above data.

3. Continuous frequency distribution (Grouped frequency distribution)

Continuous frequency distribution arises from continuous variables. i.e., from measurements done on continuous scales like height, weight, amount of power supply, etc. Hence the observations of each item falls in one and only one group (class) and the classes would be exhaustive

Example 2.3: The following table shows the frequency distribution of the test results of 50 students in Statistics course.

Test Results	Number of students
10 – 13	8
14 – 17	15
18 – 21	15
22 – 25	7
26 – 30	5
Total	50

The categories in to which the observations are distributed are called **classes** or **class intervals**. The classes should be set so that they contain all items and no two classes share the same item. This is the basic principle in the construction of such frequency distributions. We will define some concepts associated with continuous frequency distributions in the following way.

Class limits: In the above table the students are distributed in to different classes. There are 8 students with scores between 10 and 13. The numbers 10 and 13 are called lower and upper class

limits, respectively. There are 15 students with scores between 14 and 17. The numbers 14 and 17 are called lower and upper class limits respectively.

Class limits are therefore the lowest and highest values that can be included in a class. In the above examples, the numbers 10, 14, 18 and 22 are called the lower class limits (LCL) and the numbers 13, 17, 21 and 25 are called the upper class limits (UCL)

Class boundaries (real class limits): A class boundary is a number that does not appear in the stated class limits but is rather a value that *falls midway between the upper limit of one class and the lower limit of the next large one.*

In practice, the class boundaries are obtained by **adding the upper_class limit** of one class interval to *the lower limit of the next higher class* interval and dividing it by 2.

Let $d = \text{LCL of the second class} - \text{UCL of the first class}$.

Then adding $\frac{1}{2}d$ to upper limits gives the upper class boundaries (UCB) and subtracting $\frac{1}{2}d$ from the lower limits gives the lower class boundary (LCB)

For the data in the above table,

$$d = 14 - 13 = 1$$

$$\frac{1}{2}d = 0.5$$

Subtracting 0.5 from the lower Limits gives

9.5, 13.5 17.5 and 21.5 which are the lower class boundaries

Adding 0.5 to the upper limits gives 13.5, 17.5, 21.5 and 25.5 which are the upper class boundaries.

Therefore, the above table becomes

Classes	Frequency	Class boundaries
10-13	8	9.5 - 13.5
14-17	15	13.5 – 17.5
18-21	20	17.5-21.5
22-25	20	21.5-25.5
Total	50	

Or Class boundaries are obtained by subtracting half of the unit of measurement (u) from the lower limits and by adding half of u on the upper class limits of a class, where u is the distance between two possible consecutive measures. It is usually taken as 1, 0.1, 0.01, 0.001, ...

$u = 1$ if all the observations are whole numbers

$u = 0.1$ if all the observations are to one decimal places

$u = 0.01$ if all the observations to two decimal places

Let LCB= Lower class boundary

UCB= upper class boundary

Then
$$LCB_i = LCL_i - \frac{u}{2} \quad \text{and} \quad UCB_i = UCL_i + \frac{u}{2}$$

For the data in the above example, consider the 2nd class 14-17, since $u = 1$,

$$LCL_2 = 14, UCL_2 = 17, \frac{u}{2} = \frac{1}{2} = 0.5$$

$$LCB_2 = 14 - 0.5 = 13.5, UCB_2 = 17 + 0.5 = 17.5$$

Class width (class size): The size or width of a class interval is the difference between the upper and lower class boundaries and is preferred to as the class width, class size or class length

i.e.
$$CW_i = UCB_i - LCB_i$$

In the above table, for the first class, the class width is $13.5 - 9.5 = 4$ and the second class 14-17 has class width $17.5 - 13.5 = 4$. In this table all classes have **equal size which** is 4.

When all the classes are of the same size the class width can also be obtained as the difference between any two consecutive lower limits or upper limits EX: see the above table.

Class mark or class mid-point or the class interval: is a value which lies mid-way between the lower and upper limits of the class and is obtained by adding the lower and upper class limits and dividing the sum by two.

i.e.
$$\begin{aligned} \text{class mark (CM)} &= \frac{LCL + UCL}{2} \\ &= \frac{LCB + UCB}{2} \end{aligned}$$

Note that when the class size is uniform in a distribution, after finding the class mark of the first class the remaining are obtained by adding the class size. So, in the case of classes with the same size, the class width can also be obtained as the difference between any two consecutive class marks. For the distribution of the above table, the class mark of the first class is 11.5 then the class mark of the second class is $11.5 + 4 = 15.5$ the class mark of the 3rd class is $15.5 + 4 = 19.5$ and that of the fourth class is $19.5 + 4 = 23.5$ then we can have the following table.

Class Limits	Class boundaries	Class mark	Frequency
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10-13	9.5-13.5	11.5	8
14-17	13.5-17.5	15.5	15
18-21	17.5-17.5	19.5	20
22-25	21.5-25.5	23.5	7
Total			50

Basic principles for constructing a continuous frequency distribution

The basic principles or steps in constructing a continuous frequency distribution are:

1. Determine the number of classes that will be used to group the data.

The number of classes should be neither so large as to destroy the advantage of classification, nor be so small that the chief characteristic of the data is missed. The exact number of classes to use depends upon the number of figures to be classified, the size of figures, the purpose that data has to serve and the arbitrary preference of the analyst.

A small number of items to be classified justify a small number of classes. For example, if we classify 30 items into 20 classes we would lose more than we gain from the classification. If, on the other hand, we classify 15,000 items into 5 classes we would probably give away too much information.

So, in general the approximate number of classes depends upon the number of measurements and the following rough information gives us a good hint.

Sturges' Rule

To fix the number of classes (k) one can use the above method, a personal judgment depending upon the nature of investigation or decide with the help of Sturges' Rule, stating that

$$\text{Number of classes} = k = 1 + 3.22 \times \log N$$

Where N = total number of observations and log is common logarithm.

Generally, the number of classes should be between 5 and 20. That is, not less than 5 and not greater than 20 classes should be used for any kind of distribution.

2. The size of the class has to be determined.

Whenever possible, all classes should be of the same size. This facilitates the analysis of the data and simplifies comparison between different classes.

A frequency distribution with equal class size can be presented pictorially with greater ease.

However, in some cases equal size is either impossible or undesirable.

If the number of classes is known and if it is decided to use classes of equal size, the determination of the size is relatively simple. Since the class size depends upon the number of classes and the extent to which the values of the variable are spread or dispersed, the following simple formula can be used.

$$\text{Class width}(cw) \cong \frac{\text{Range}}{\text{Number of classes}} \text{ or } cw \cong \frac{R}{k}$$

Where $R = \text{Range} = \text{Highest Value} - \text{Smallest Value}$

3. Determine the lower class limit of the first class so that the smallest item falls in this class. The remaining lower class limits are obtained using the following relations.

$$LCL_2 = LCL_1 + cw, LCL_3 = LCL_2 + cw, LCL_4 = LCL_3 + cw, \dots, LCL_{i+1} = LCL_i + cw$$

4. Determine the upper class limit of the first class using the formula

$UCL_1 = LCL_1 + cw - u$. The remaining lower class limits are obtained using the following relations.

$$UCL_2 = UCL_1 + cw, UCL_3 = UCL_2 + cw, \dots, UCL_{i+1} = UCL_i + cw$$

5. Complete the continuous frequency distribution with the respective class frequencies.

Example 2.4: Following are marks of (out of 100) obtained by 50 students in Statistics.

41 50 69 77 88 92 40 51 67 75 87 94 93 86 72 62 53 49 57 67
 70 85 97 95 83 79 68 52 44 44 55 64 75 83 74 60 56 42 56 69
 70 42 64 52 63 60 59 61 65 78

- a. Construct a continuous frequency distribution with suitable number of classes using Sturges' Rule.
- b. Complete the distribution obtained in (a) with the class boundaries and class marks.

4. Relative Frequency Distribution

The relative frequency of a class shows the relative concentration of items in a given class interval to the other classes of a frequency distribution.

$$\text{Relative frequency of a class} = \frac{\text{class frequency}}{\text{Total frequency}}$$

Relative frequencies are usually given as decimals or percentages.

Example 2.5: The following table shows an example of relative frequency distribution.

Wages (X)	Number of Workers(f)	Relative frequency	
		In decimals	In %
75-80	9	0.09	9%

80-85	12	0.12	12%
85-90	15	0.15	15%
90-95	11	0.11	11%
95-100	20	0.2	20%
100-105	20	0.2	20%
105-110	11	0.11	11%
110-115	2	0.02	2%
Total	100	1.00	100%

5. Cumulative Frequency Distribution

The cumulative frequency of value of a variable (a class) is the sum of all the frequencies preceding or succeeding that value (class) including the frequency of that value (class) there are two types of cumulative frequency distributions namely the “less than” cumulative frequency distribution and the “more than” cumulative frequency distribution.

a) “Less than” cumulative frequency distribution

Less than cumulative frequency for any value of the variable (or class) is obtained by adding values (or classes), including the frequency of that value (class) against which the totals are written, provided the values (Classes) are arranged in ascending order of magnitude. Or for grouped frequency distribution it is the sum of all frequencies lying below the upper class boundaries of each class.

Example 2.6: The table below shows the ‘less than’ cumulative frequency distribution of marks of 70 students in a class.

Marks	Frequency	‘Less than’ Cumulative Frequency
30-35	5	5
35-40	10	5+10=15
40-45	15	15+15=30
45-50	30	30+30=60
50-55	5	60+5=65

55-60	5	$65+5=70$
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The above 'less than' cumulative frequency distribution can also be written as follows

Marks	Frequency
Less than 30	0
Less than 35	5
Less than 40	15
Less than 45	30
Less than 50	60
Less than 55	65
Less than 60	70

b) 'more than' cumulative frequency distribution

The 'more than' cumulative frequency is obtained similarly by finding the cumulative totals of frequencies starting from the highest value of the variable (class) to the lowest value (class). Thus in the above illustration the number of students with marks 'more than 50' is $5+5=10$, and 'more than 40' is $15+30+5+5=55$ and so on. The complete 'more than' type cumulative frequency distribution for this data is given below:

'More than' cumulative frequency distribution of marks of 70 students

Marks	Frequency	'more than' cumulative frequency
30-35	5	$65+5=70$
35-40	10	$55+10=65$
40-45	15	$40+15=55$
45-50	30	$10+30=40$
50-55	5	$5+5=10$
55-60	5	5

The above 'more than' C.F. Distribution can also be expressed in the following form:

Marks	Number of students
More than 30	70

More than 35	65
More than 40	55
More than 45	40
More than 50	10
More than 55	5
More than 60	0

Remark: In ‘less than’ C.F. Distribution , the c.f. refers to the upper class boundary of the corresponding class and in ‘more than’ cumulative frequency distribution, the c.f. refers to the lower class boundary of the corresponding class

Exercise: Convert the following distribution in to ‘more than’ frequency distribution

Weekly wages less than birr	Number of workers
20	41
40	92
60	156
80	194
100	201

2.4 DIAGRAMMATIC AND GRAPHICAL PRESENTATION OF DATA

Dear students, can you mention some diagrammatic and graphical methods of data presentation? Do we use the same diagram and graph for different purposes or do we have different diagrams and graphs for different purposes? If know them, compare with what is discussed below. We will discuss them in this section.

2.4.1 DIAGRAMMATIC PRESENTATION OF DATA

When the basis of classification is not quantitative, i.e.; when the data are of attribute nature, statistical data can be presented diagrammatically using charts. The charts could be bar chart, pie-chart or picot gram all of which having specific uses depending upon the nature of the information to be depicted.

2.4.1.1. BAR CHARTS

Bar charts are drawn almost in the same way as graphs. Data are presented by a series of bars, the heights of each bar showing the size of the observation represented.

While drawing the bar charts:

- I. The width of the bar should be kept uniform and
- II. The gaps between successive bars should remain the same

There are four main types of bar charts serving different purposes. These are simple bar charts, component charts, percentage component bar charts and multiple bar charts.

Simple bar chart

In simple bar charts, each bar represents one and only one figure. A simple bar chart is usually constructed to represent total only.

Example 2.7: the following shows the number of students attending in four departments.

Department	Mathematics	Statistics	Physics	Chemistry
Number of student	56	45	40	50

Construct a simple bar chart for the above table.

Component (sub-divided) bar chart

The component bar chart gives the break up in parts which constitutes the aggregate in a year, place or sector. In such type of chart, it is possible to compare changes in part, in aggregate, as well.

Multiple bar charts

Here, the interrelated components are shown as adjoining bars. Colored or marked differently that allowing comparison between different parts.

Example 2.8: The table and chart below show the revenue, expenditure of a country on education

Education	Expenditure (in million)		
	1978-80	1980-81	1981-82
Primary	60	80	40
Secondary	40	60	60
Higher Education	20	40	20
Total	120	180	120

Construct a component and multiple bar chart for the above table.

Pie-chart (angular chart)

A pie-chart is a circle divided by radial lines into sections or slices so that the area of each section is proportional to the size of the amount represented. It is a simple descriptive display of data that sum

to a given total. A pie-chart is probably the most illustrative way of displaying quantities as percentage of a given total. The total area of the pie represents 100 percent of the quantity of interest (the sum of the variable values in all categories of the slice denotes).

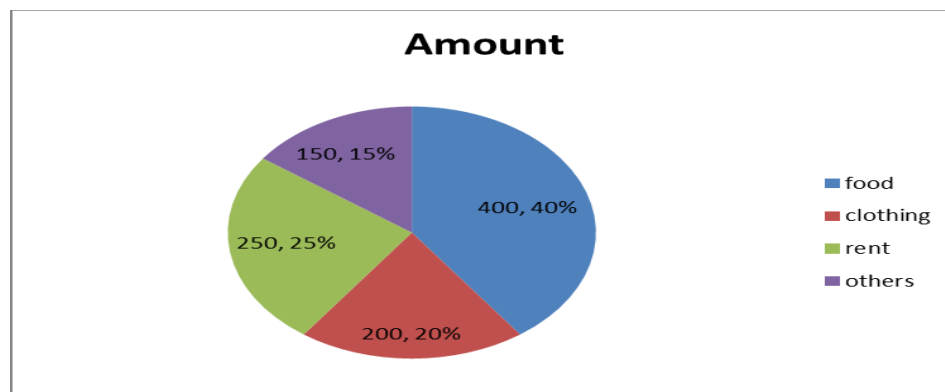
Thus, a pie-chart indicates relative frequencies by slicing up a circle into distinct sectors.

The sum of angles at a point being 360° , the component parts of the data are expressed as proportions of 360° and the sectors of circle represent these parts. The degrees corresponding to components are obtained by dividing the amount for each item divided by the total and multiplying by 360° and to be drawn by means of a protractor.

In order to draw pie chart, it is convenient to form beforehand a table of percentages and the corresponding angles to be drawn at the center of the circle.

Example 2.10: The following table shows the monthly expense of family with income of 1000 Birr. Construct pie chart.

Item	Food	Clothing	Rent	Others	Total
Amount(in Birr)	400	200	250	150	1000



2.4.2 GRAPHICAL PRESENTATION

After organizing data in to frequency distributions it is often helpful to present data graphically. Graphs communicate the essential characteristics of a frequency distribution in pictorial form so that one can readily identify these characteristics and can compare one frequency distribution with another.

Histogram, frequency polygon and cumulative frequency curves are common ways of representing frequency distribution graphically.

Histogram

A histogram is a graphical display of the distribution of a data set. A histogram looks like a vertical bar graph, except that the columns touch each other.

The given grouped data is plotted in the form of a series of rectangles. Class boundaries are marked along the x-axis and the frequencies along the y- axis according to a suitable scale. If all the classes are of the same size, the height of the rectangles can be taken to be numerically equal to the class frequencies.

If on the other hand the size of the class intervals is not uniform, the height of the rectangles can be adjusted by taking the “frequency density” of the corresponding classes as scale for the vertical axis.

$$\text{Frequency density} = \frac{\text{class Frequency}}{\text{Class width}}$$

A histogram gives us an idea about the shape of the data distribution. It can indicate to us, graphically, where the center of the data distribution lies. It will also reveal whether the distribution is symmetric or skewed.

Example 2.11: Construct a histogram for the following frequency distribution

Class limits	f_i	Class boundary
10-19	4	9.5-19.5
20-29	5	19.5-29.5
30-39	8	29.5-39.5
40-49	6	39.5-49.5
50-59	2	49.5-59.5

Frequency

polygon

Is a line chart of frequency distribution in which either the values of discrete variables or the class marks of classes are plotted against the frequencies and these plotted points are joined together by straight lines.

It is thus a graphic presentation tool that may be used as an alternative to the histogram. For a large number of classes a frequency polygon is preferable.

For a frequency distribution where class intervals are equal, the area of frequency polygon is equal to the area of the histogram.

Cumulative frequency curve (ogive) - It is the graphic representation of cumulative frequency distribution. The ogive curve can be traced either on less than basis or more than basis.

- a) **‘Less than Ogive’**: Upper class boundaries are plotted against the ‘less than’ cumulative frequencies.
- b) **‘More than’ Ogive**: Lower class boundaries are plotted against the ‘more than’ cumulative frequencies.

Example2.13: Construct (a) The ‘Less than’ ogive and

(b) The ‘More than’ ogive for the above frequency distribution.